NDM

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This is a Jupyter Book.

Check out the Jupyter Book documentation for more information.

Contents

- *Overview*
- *NDM and LAMMPS*
- *Prerequisites*
- *NDM HowTo*
- *Create jupyter-book_howto*

CHAPTER

ONE

OVERVIEW

- NDM TODO

etc.

CHAPTER

TWO

NDM AND LAMMPS

TODO traduction and formattage

```
NDM peut utiliser lammps comme subroutine de forces et contraintes.

utilise liblammps_serial.a

compilation de NDM avec make LAMMPS_VERSION=1 (voir les ifdef correspondants)
crée rundm90_lammps

Utilisation de rundm90_lammps :
1/dans .din :
ipotentiel =-10 potentiel lammps sans charge variable
OU
ipotentiel =-11 potentiel lammps à charge variable
Ce choix joue sur les fichiers data et sur les fichiers potentiels

Préciser units_lammps (e.g. units_lammps='metal') définit les changements d'unités␣
↪entre lammps et NDM
2/potentiel dans NDM : simple.potin
exemple ipotentiel =-10
1 ! nb de types
6.0 rue
55.84700 26 'Fe'
exemple ipotentiel =-11 (avec la charge initiale en plus)
 2 ! nb de types
11.0 rue
15.9994 8.0 'O ' -1.613626              ! CM, numero atomique(O), ty, Q
238.03  92.0 'U'  3.227252

3/NDM va créer un fichier conf.lmp avec la configuration de départ à partir de .gin␣
↪ou .cin

4/ IL faut avoirun fichier in.lammps avec la description minimale de la configuration␣
↪de lammps. Exemple:
units                metal
atom_style      atomic
atom_modify     map array
box tilt large
read_data        conf.lmp
mass       1  55.84500
pair_style        eam/alloy
pair_coeff      * *        M07_eam.fs Fe

5/ in .lammps contient le nom du fichier de potentiel (ici M07_eam.fs ) qui doit être␣
↪présent.
```

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```
Le code a été testé pour des calculs NVE et NPH en eam et SMTBQ. Les changements de␣
↪repères entre NDM et LAMMPS sont gérés automatiquement. Pas besoin que la boite NDM␣
↪soit conforme au format lammps.

Programmation :
Ca repose sur des appels de lammps dans NDM, à coup de call lammps-open_..., call␣
↪lammps_gather, etc. voir ndm_lammps.F90.
N.B. Pour les contraintes , on a été obligé de coder en dur la dimension du tableau␣
↪des contraintes dans LAMMPS.F90:
    if (id=="thermo_press") then
       Cvectorsize=6
       vectorsize = int(Cvectorsize, kind(vectorsize))
       return
    endif
    Ca n'est pas glorieux mais ça marche...
```

CHAPTER

THREE

PREREQUISITES

Prerequisite are *only* a PYTHON-3 environment.

- Using Miniconda. *It is easy*.
- Mandatory prerequisites python3, PyQt5, numpy, pandas, jupyter etc.
- See List python environment configuration

Example of activate a miniconda environment on is246206:

```
# set conda in $PATH etc.
source ${HOME}/commons/conda_initialize_is246206

conda env list
  # conda environments:
  #
  base                  *  /export/home/catA/miniconda
  py3qt5                   /export/home/catA/miniconda/envs/py3qt5
  pypinns                  /export/home/catA/miniconda/envs/pypinns

conda activate py3qt5

which python
  /export/home/catA/miniconda/envs/py3qt5/bin/python

envs -e PATH
  PATH                           = /export/home/catA/miniconda/envs/py3qt5/bin
                                   /export/home/catA/miniconda/condabin
                                   .
                                   /home/catA/wambeke
                                   /home/catA/wambeke/commons
```

3.1 Execute an howto example

- Do that after reading below chapter for best comprehension.
- Directly (**outside jupyter**) CLI execution example, **for simple test**.

```
cd .../PACKAGESPY/doc

python
>> import PACKAGESPY_howto_level_2 as HT2
>> HT2.test_2()
```

3.2 Some MINICONDA tricks

Here for memory.

3.2.1 List python configurarion

For example.

```
conda info

     active environment : py3qt5
    active env location : /export/home/catA/miniconda/envs/py3qt5
            shell level : 2
       user config file : /home/catA/wambeke/.condarc
 populated config files :
          conda version : 22.9.0
    conda-build version : not installed
         python version : 3.9.12.final.0
       virtual packages : __cuda=11.6=0
                          __linux=5.17.12=0
                          __glibc=2.33=0
                          __unix=0=0
                          __archspec=1=x86_64
       base environment : /export/home/catA/miniconda  (writable)
      conda av data dir : /export/home/catA/miniconda/etc/conda
  conda av metadata url : None
           channel URLs : https://repo.anaconda.com/pkgs/main/linux-64
                          https://repo.anaconda.com/pkgs/main/noarch
                          https://repo.anaconda.com/pkgs/r/linux-64
                          https://repo.anaconda.com/pkgs/r/noarch
          package cache : /export/home/catA/miniconda/pkgs
                          /home/catA/wambeke/.conda/pkgs
       envs directories : /export/home/catA/miniconda/envs
                          /home/catA/wambeke/.conda/envs
               platform : linux-64
             user-agent : conda/22.9.0 requests/2.28.1 CPython/3.9.12 Linux/5.17.12-
↪100.fc34.x86_64 fedora/34 glibc/2.33
                UID:GID : 94740:470572
             netrc file : None
           offline mode : False
```

3.2.2 List python environment configuration

For example.

```
conda activate py3qt5
conda env export > ./doc/conda_environment_py3qt5.yaml
cat ./doc/conda_environment_py3qt5.yaml
  name: py3qt5
  channels:
    - conda-forge
    - defaults
```

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```
dependencies:
  - _libgcc_mutex=0.1=conda_forge
  - _openmp_mutex=4.5=2_kmp_llvm
  - argon2-cffi=21.3.0=pyhd3eb1b0_0
  - argon2-cffi-bindings=21.2.0=py39h7f8727e_0
  - asttokens=2.0.5=pyhd3eb1b0_0
  - attrs=21.4.0=pyhd3eb1b0_0
  - backcall=0.2.0=pyhd3eb1b0_0
  - beautifulsoup4=4.11.1=py39h06a4308_0
  - blas=1.0=mkl
  - bleach=4.1.0=pyhd3eb1b0_0
  - bottleneck=1.3.4=py39hce1f21e_0
  - brotli=1.0.9=he6710b0_2
  - brotlipy=0.7.0=py39h27cfd23_1003
  - ca-certificates=2022.07.19=h06a4308_0
  - certifi=2022.9.24=py39h06a4308_0
  - cffi=1.15.0=py39hd667e15_1
  - charset-normalizer=2.0.4=pyhd3eb1b0_0
  - cryptography=37.0.1=py39h9ce1e76_0
  - cycler=0.11.0=pyhd3eb1b0_0
  - cython=0.29.28=py39h295c915_0
  - dbus=1.13.18=hb2f20db_0
  - debugpy=1.5.1=py39h295c915_0
  - decorator=5.1.1=pyhd3eb1b0_0
  - defusedxml=0.7.1=pyhd3eb1b0_0
  - distro=1.5.0=pyhd3eb1b0_1
  - entrypoints=0.4=py39h06a4308_0
  - executing=0.8.3=pyhd3eb1b0_0
  - expat=2.4.4=h295c915_0
  - fontconfig=2.13.1=h6c09931_0
  - fonttools=4.25.0=pyhd3eb1b0_0
  - freetype=2.11.0=h70c0345_0
  - giflib=5.2.1=h7b6447c_0
  - glib=2.69.1=h4ff587b_1
  - gst-plugins-base=1.14.0=h8213a91_2
  - gstreamer=1.14.0=h28cd5cc_2
  - icu=58.2=he6710b0_3
  - idna=3.3=pyhd3eb1b0_0
  - intel-openmp=2021.4.0=h06a4308_3561
  - ipykernel=6.9.1=py39h06a4308_0
  - ipython=8.3.0=py39h06a4308_0
  - ipython_genutils=0.2.0=pyhd3eb1b0_1
  - ipywidgets=7.6.5=pyhd3eb1b0_1
  - jedi=0.18.1=py39h06a4308_1
  - jinja2=3.0.3=pyhd3eb1b0_0
  - jpeg=9e=h7f8727e_0
  - jupyter=1.0.0=py39h06a4308_7
  - jupyter_client=7.2.2=py39h06a4308_0
  - jupyter_console=6.4.3=pyhd3eb1b0_0
  - jupyter_core=4.10.0=py39h06a4308_0
  - jupyterlab_pygments=0.1.2=py_0
  - jupyterlab_widgets=1.0.0=pyhd3eb1b0_1
  - kiwisolver=1.4.2=py39h295c915_0
  - lcms2=2.12=h3be6417_0
  - ld_impl_linux-64=2.38=h1181459_1
  - libffi=3.3=he6710b0_2
```

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```
  - libgcc-ng=12.1.0=h8d9b700_16
  - libpng=1.6.37=hbc83047_0
  - libsodium=1.0.18=h7b6447c_0
  - libstdcxx-ng=12.1.0=ha89aaad_16
  - libtiff=4.2.0=h2818925_1
  - libuuid=1.0.3=h7f8727e_2
  - libwebp=1.2.2=h55f646e_0
  - libwebp-base=1.2.2=h7f8727e_0
  - libxcb=1.15=h7f8727e_0
  - libxml2=2.9.14=h74e7548_0
  - libxslt=1.1.35=h4e12654_0
  - libzlib=1.2.12=h166bdaf_2
  - llvm-openmp=14.0.4=he0ac6c6_0
  - lxml=4.8.0=py39h1f438cf_0
  - lz4-c=1.9.3=h295c915_1
  - markupsafe=2.1.1=py39h7f8727e_0
  - matplotlib=3.5.1=py39h06a4308_1
  - matplotlib-base=3.5.1=py39ha18d171_1
  - matplotlib-inline=0.1.2=pyhd3eb1b0_2
  - mistune=0.8.4=py39h27cfd23_1000
  - mkl=2021.4.0=h06a4308_640
  - mkl-service=2.4.0=py39h7f8727e_0
  - mkl_fft=1.3.1=py39hd3c417c_0
  - mkl_random=1.2.2=py39h51133e4_0
  - munkres=1.1.4=py_0
  - nbclient=0.5.13=py39h06a4308_0
  - nbconvert=6.4.4=py39h06a4308_0
  - nbformat=5.3.0=py39h06a4308_0
  - ncurses=6.3=h7f8727e_2
  - nest-asyncio=1.5.5=py39h06a4308_0
  - notebook=6.4.11=py39h06a4308_0
  - numexpr=2.8.1=py39h807cd23_2
  - numpy=1.22.3=py39he7a7128_0
  - numpy-base=1.22.3=py39hf524024_0
  - openssl=1.1.1q=h7f8727e_0
  - packaging=21.3=pyhd3eb1b0_0
  - pandas=1.4.2=py39h295c915_0
  - pandas-datareader=0.10.0=pyhd3eb1b0_0
  - pandocfilters=1.5.0=pyhd3eb1b0_0
  - parso=0.8.3=pyhd3eb1b0_0
  - pcre=8.45=h295c915_0
  - pexpect=4.8.0=pyhd3eb1b0_3
  - pickleshare=0.7.5=pyhd3eb1b0_1003
  - pillow=9.0.1=py39h22f2fdc_0
  - pip=21.2.4=py39h06a4308_0
  - ply=3.11=py_1
  - prometheus_client=0.13.1=pyhd3eb1b0_0
  - prompt-toolkit=3.0.20=pyhd3eb1b0_0
  - prompt_toolkit=3.0.20=hd3eb1b0_0
  - ptyprocess=0.7.0=pyhd3eb1b0_2
  - pure_eval=0.2.2=pyhd3eb1b0_0
  - pycparser=2.21=pyhd3eb1b0_0
  - pygments=2.11.2=pyhd3eb1b0_0
  - pyomo=6.4.2=py39h5a03fae_0
  - pyopenssl=22.0.0=pyhd3eb1b0_0
  - pyparsing=3.0.4=pyhd3eb1b0_0
```

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```
  - pyqt=5.9.2=py39h2531618_6
  - pyrsistent=0.18.0=py39heee7806_0
  - pysocks=1.7.1=py39h06a4308_0
  - python=3.9.12=h12debd9_1
  - python-dateutil=2.8.2=pyhd3eb1b0_0
  - python-fastjsonschema=2.15.1=pyhd3eb1b0_0
  - python_abi=3.9=2_cp39
  - pytz=2022.1=py39h06a4308_0
  - pyyaml=6.0=py39h7f8727e_1
  - pyzmq=22.3.0=py39h295c915_2
  - qt=5.9.7=h5867ecd_1
  - qtconsole=5.3.0=pyhd3eb1b0_0
  - qtpy=2.0.1=pyhd3eb1b0_0
  - readline=8.1.2=h7f8727e_1
  - requests=2.28.0=py39h06a4308_0
  - send2trash=1.8.0=pyhd3eb1b0_1
  - setuptools=61.2.0=py39h06a4308_0
  - sip=4.19.13=py39h295c915_0
  - six=1.16.0=pyhd3eb1b0_1
  - soupsieve=2.3.1=pyhd3eb1b0_0
  - sqlite=3.38.5=hc218d9a_0
  - stack_data=0.2.0=pyhd3eb1b0_0
  - terminado=0.13.1=py39h06a4308_0
  - testpath=0.6.0=py39h06a4308_0
  - tk=8.6.12=h1ccaba5_0
  - tornado=6.1=py39h27cfd23_0
  - traitlets=5.1.1=pyhd3eb1b0_0
  - typing-extensions=4.1.1=hd3eb1b0_0
  - typing_extensions=4.1.1=pyh06a4308_0
  - tzdata=2022a=hda174b7_0
  - urllib3=1.26.9=py39h06a4308_0
  - wcwidth=0.2.5=pyhd3eb1b0_0
  - webencodings=0.5.1=py39h06a4308_1
  - wheel=0.37.1=pyhd3eb1b0_0
  - widgetsnbextension=3.5.2=py39h06a4308_0
  - xz=5.2.5=h7f8727e_1
  - yaml=0.2.5=h7b6447c_0
  - zeromq=4.3.4=h2531618_0
  - zlib=1.2.12=h7f8727e_2
  - zstd=1.5.2=ha4553b6_0
  - pip:
    - alabaster==0.7.12
    - anyio==3.6.1
    - babel==2.10.3
    - click==8.1.3
    - colorama==0.4.5
    - docutils==0.17.1
    - gitdb==4.0.9
    - gitpython==3.1.27
    - greenlet==1.1.3
    - imagesize==1.4.1
    - importlib-metadata==4.12.0
    - jsonschema==3.2.0
    - jupyter-book==0.13.1
    - jupyter-cache==0.4.3
    - jupyter-server==1.18.1
```

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```
    - jupyter-server-mathjax==0.2.6
    - jupyter-sphinx==0.3.2
    - latexcodec==2.0.1
    - linkify-it-py==1.0.3
    - markdown-it-py==1.1.0
    - mdit-py-plugins==0.2.8
    - myst-nb==0.13.2
    - myst-parser==0.15.2
    - nbdime==3.1.1
    - psutil==5.9.1
    - pybtex==0.24.0
    - pybtex-docutils==1.0.2
    - pydata-sphinx-theme==0.8.1
    - smmap==5.0.0
    - sniffio==1.2.0
    - snowballstemmer==2.2.0
    - sphinx==4.5.0
    - sphinx-book-theme==0.3.3
    - sphinx-comments==0.0.3
    - sphinx-copybutton==0.5.0
    - sphinx-design==0.1.0
    - sphinx-external-toc==0.2.4
    - sphinx-jupyterbook-latex==0.4.6
    - sphinx-multitoc-numbering==0.1.3
    - sphinx-thebe==0.1.2
    - sphinx-togglebutton==0.3.2
    - sphinxcontrib-applehelp==1.0.2
    - sphinxcontrib-bibtex==2.5.0
    - sphinxcontrib-devhelp==1.0.2
    - sphinxcontrib-htmlhelp==2.0.0
    - sphinxcontrib-jsmath==1.0.1
    - sphinxcontrib-qthelp==1.0.3
    - sphinxcontrib-serializinghtml==1.1.5
    - sqlalchemy==1.4.40
    - uc-micro-py==1.0.1
    - websocket-client==1.4.0
    - zipp==3.8.1
prefix: /export/home/catA/miniconda/envs/py3qt5
```

CHAPTER FOUR

NDM HOWTO

- Here, you need to know elementary python scripting, **obviously**.
- To create your interactive documentation, may be you have to write python script to make examples etc.
- This could be easy, but **not** without understanding.
- For details, see current python script example NDM_howto.py

4.1 Usage by python scripting

- You may use/modify the next cell

```
### Usage by python scripting

# example of create model tree from scratch with all default branches and leaves␣
↪values
print('hello world')
```

```
hello world
```

CHAPTER

FIVE

CREATE JUPYTER-BOOK_HOWTO

- Create `.ipynb` files as documentation **and** execution (… experimentation).
- Example on is246206, change SANDBOX repository *at yours needs*.

```
export MYROOT=/export/home/catA/wambeke/NDM_ROODIR
alias cd_doc="cd ${MYROOT}/NDM/docs/doc_2023"

f_init_conda
conda activate py3qt5
conda list | grep jupyter-book
which jupyter-notebook
which jupyter-book

cd_doc
jupyter-notebook
```

- **… and now you make .ipynb files (as NDM_howto.ipynb etc…) at yours needs**
- **… and after you copy yours .ipynb files etc… to NDM/docs/…, and modify docs/…/_toc.yml**

CHAPTER SIX

CREATE DOCUMENTATION HTML PLUS PDF WITH JUPYTER-BOOK

- On is246206

6.1 Prerequisites jupyter-book

```
pip install jupyter-book

sudo dnf remove texlive
sudo dnf install texlive-scheme-full

# https://unix.stackexchange.com/questions/199638/how-to-fully-install-latex-in-fedora
# sudo dnf install latexmk
# sudo dnf install texlive-pgfplots
# sudo dnf install texlive-bbm
```

6.2 Create doc html plus pdf

```
cd_doc

# to get first example _config.yml and _toc.yml
# jupyter-book create example_tmp
# cp example_tmp._* .
# rm -rf example_tmp

export BOOK=$(pwd)              # as first empty example

# jupyter-book create ${BOOK}
tree ${BOOK}
rm -rf ${BOOK}/_build ${BOOK}/_tmp ${BOOK}/__pycache__

# html
jupyter-book build ${BOOK}
firefox ${BOOK}/_build/html/index.html

# pdf OK
# using CFI.colorize_file_to_markdown("PACKAGESPY_howto.py") etc.
jupyter-book build ${BOOK} --builder pdflatex
```

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```
evince ${BOOK}/_build/latex/PACKAGESPY.pdf
cp ${BOOK}/_build/latex/PACKAGESPY.pdf ${BOOK}/.
```